

Transformers Know When They Don't Know: Layer-Wise Distance Awareness for OOD Detection

1. Baselines overview

1.1. ODIN [1]

Summary: Temperature scaling and adding small perturbations to the input can separate the softmax score distributions between in- and out-of-distribution images.

Input Preprocessing by adding small perturbations to it. The preprocessed image is given by:

$$x' = x - \varepsilon \cdot \text{sign}(-\nabla_x \log(S_{y_{best}}(x; T)))$$
$$S_i(x; T) = \frac{\exp(f_i(x)/T)}{\sum_{j=1}^N \exp(f_j(x)/T)}$$

aim to increase the softmax score.

Hyper-parameters: For each out-of-distribution dataset, we randomly hold out 1,000 images for tuning the parameters T and ε . For temperature T , we select among 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000; and for perturbation magnitude ε we choose from 21 evenly spaced numbers starting from 0 and ending at 0.004.

Steps:

- Hyper-parameters: Following VOS [4] (result report in SAFE [5] paper), $T=1000$, $\varepsilon=0$.
- Since $\varepsilon=0$, we simply feed the image x into the neural network, calculate its softmax score $S_{y_{best}}(x; T)$.

We say that the image x is an in-distribution example if the softmax score is above the threshold and that the image x is an out-of-distribution example, otherwise.

1.2. Energy Score [3]

Summary: Energy scores are theoretically aligned with the probability density of the inputs and are less susceptible to the overconfidence issue. Two types: postdocs and retraining (fine-tune the model).

Steps:

- We adopt the postdoctoral approach, as we aim to avoid modifying the model. Follow [3], we set temperature = 1.
- Compute score:

$$E(x; f) = -T \cdot \log \sum_i^K e^{f_i(x)/T}, \quad K \text{ is the number of classes.}$$

Lower energies to the in-distribution data, and higher energies to the OOD data.

References

1. Enhancing The Reliability of Out-of-distribution Image Detection in Neural Networks. ICLR 2018.
2. A Simple Unified Framework for Detecting Out-of-Distribution Samples and Adversarial Attacks. NeurIPS 2018.

3. Energy-based Out-of-distribution Detection. NeurIPS 2020.
4. VOS: Learning What You Don't Know by Virtual Outlier Synthesis.
5. SAFE: Sensitivity-Aware Features for Out-of-Distribution Object Detection.